Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the air starter to prevent accidental engine starting. Refer to Procedure 012-022 in Section 12. (/qs3/pubsys2/xml/en/procedures/102/102-012-022.html)
- Remove the transmission, clutch, and all related components. See the equipment manufacturer service

information.

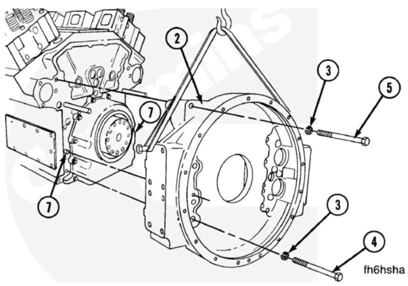
- Remove the engine speed sensor, use the special socket tool, Part Number 4918623, or equivalent. Refer to Procedure 019-042 in the Troubleshooting and Repair Manual Electronic Control System, KTA38GC CM558, Bulletin 4021665. (/qs3/pubsys2/xml/en/procedures/122/122-019-042.html)
- Drain the engine oil. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Remove the starter motor. Refer to Procedure 013-020 in Section 13. (/qs3/pubsys2/xml/en/procedures/28/28-013-020-tr.html)
- Remove the oil pan adapter assembly. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)

Remove

Note : Newer engines have a flywheel housing that also contains a rear crankshaft seal. When this housing is used, the rear seal housing or the rear gear drive housing is not used. The newer housing is referred to as “Housing with Rear Seal” in the procedure. Older housings are referred to as “Housing without Rear Seal”.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



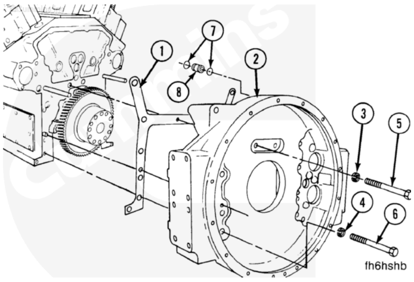
Flywheel Housing without Rear Seal

- On engines equipped with flywheel housings without a rear seal, remove the capscrews (4) and (5). There are twelve capscrews.
- Check the grade of the capscrews.

- If the capscrews are Society of Automotive Engineers (SAE) Grade 5, discard them and replace with SAE Grade 8 capscrews.
- If the capscrews are SAE Grade 8, retain them for installation.
- After removing the first few capscrews, install guide bolts to aid in the disassembly procedure.
- Remove the flywheel housing.
- The dowel pins do **not** have to be removed unless they are damaged.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Flywheel Housing with Rear Seal

- On housings with rear seals, remove the capscrews (5) and (6). There are twelve capscrews.
- Check the grade of the capscrews.
- If the capscrews are SAE Grade 5, discard them and replace with SAE Grade 8 capscrews.
- If the capscrews are SAE Grade 8, retain them for installation.
- After removing the first few capscrews, install guide bolts to aid in the disassembly procedure.
- Remove the flywheel housing.
- The dowel pins do **not** have to be removed unless they are damaged.
- Remove the rear seal.
- Discard the gasket.

Clean and Inspect for Reuse

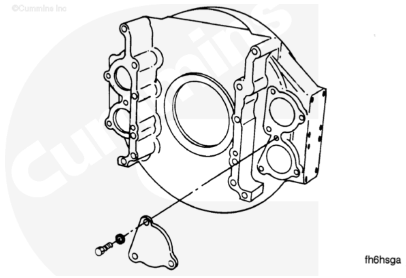
⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the risk of personal

injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam to clean the flywheel housing.

Dry with compressed air.

Inspect all surfaces for nicks, burrs, or cracks.

Use a fine crocus cloth to remove small nicks and burrs.

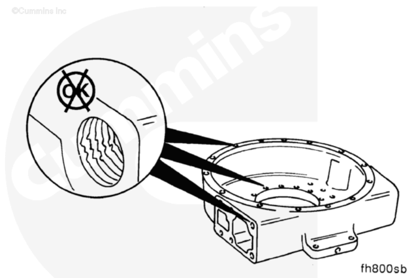
Install the covers.

Note : Gaskets are only required on wet type flywheel housings.

Inspect all threaded capscrew holes for damage.

Inspect the engine speed sensor threaded hole for damage.

Repair or replace the housing if the threaded holes are damaged.

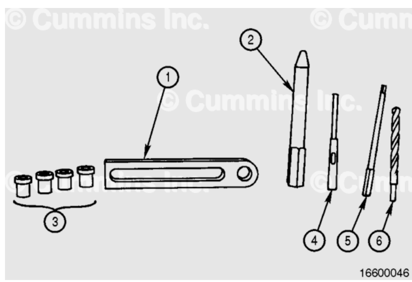


Redowel

The tools needed to perform this procedure are:

Part Number ST-1232, Drill Ream Fixture, that contains:

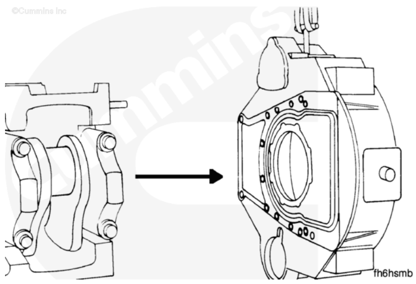
1. Plate, Part Number ST-1232-1
2. Locator pin, Part Number 3375052
3. Drill/Ream actual sizes. Bushing set depends on the dowel size



4. Drill adapter locally obtained; use to adapt open-shank reamers to drill-chuck
5. Reamer locally obtained
6. Drill bit locally obtained.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

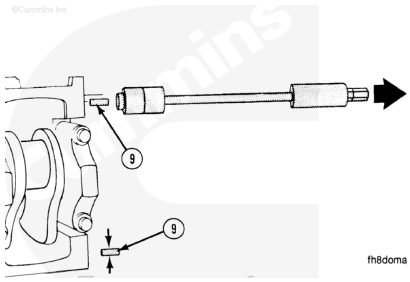


Note : The procedure remains the same, even though the following illustrations show a K19 flywheel housing.

Remove the flywheel housing.

Use a dowel pin extractor, Part Number 3163720, or equivalent. Remove the two dowels (9) from the block.

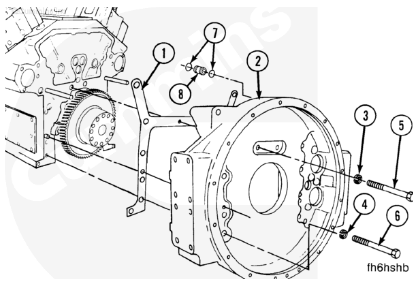
Measure a dowel pin that is removed so that an oversize dowel pin size can be determined.



Install the flywheel housing.

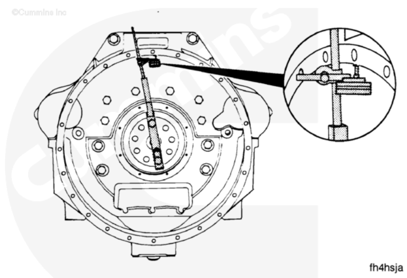
The capscrews (5) and (6) **must** be SAE Grade 8.

Do **not** tighten the capscrews. The flywheel housing will need to be aligned.



Align the flywheel housing to the crankshaft.

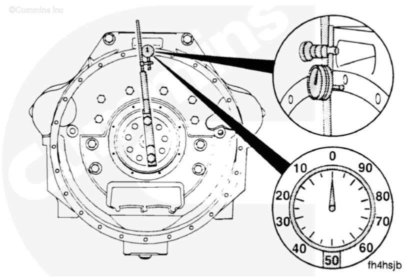
Move the housing with a mallet until the bore is within specifications. Make sure that the bore and face of the housing are in alignment.



Note : Unless tooling denotes otherwise, when measuring the flywheel housing run out subtract 0.15 mm [0.006 in] from the run out measurement to account for the crankshaft bearing clearance.

Flywheel Housing - Radial Run Out				
Inside Diameter		SAE Number	Maximum Runout	
mm	in		mm	in
787	31.0	00	0.48	0.016
648	25.5	0	0.41	0.019

Flywheel Housing - Face Run Out				
Inside Diameter		SAE Number	Maximum Runout	
mm	in		mm	in
787	31.0	00	0.48	0.016
648	25.5	0	0.41	0.019



When the housing is in alignment, tighten the capscrews in the sequence shown.

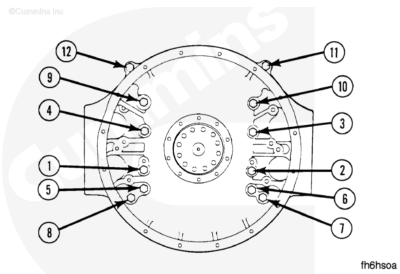
Flywheel Housing - Torque Sequence

Torque Value:

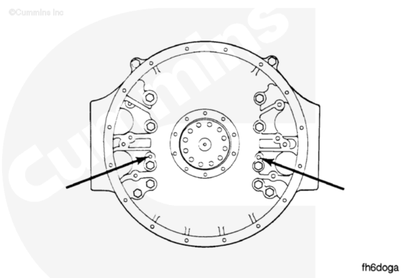
¾ - 10 Inch Capscrews

1. 150 n•m [110 ft-lb]
2. 305 n•m [225 ft-lb]
3. 460 n•m [340 ft-lb]

After the capscrews are tightened, check the alignment again.

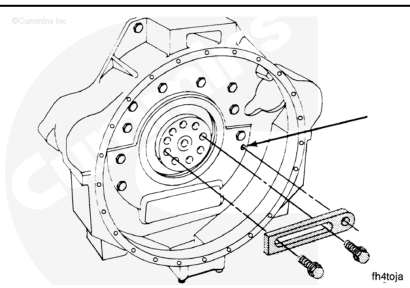


This illustration shows the location of the dowel holes on the flywheel housing.



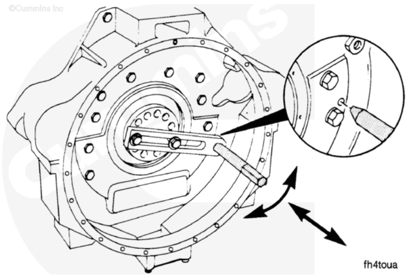
Use appropriate size capscrews. Attach Part Number ST-1232-1, Plate, that is contained in Part Number ST-1232 Drill Ream Fixture, to the crankshaft.

Do **not** tighten the capscrews so much that the plate does **not** move.



⚠ CAUTION ⚠

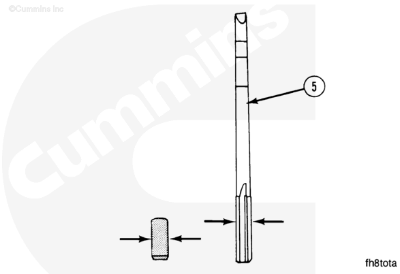
The crankshaft must be "locked" in position. It can turn during reaming.



Use the locator pin to align the plate with the hole for the dowel pin. Tighten the capscrews. The taper on the pin **must** engage the dowel pin hole.

The locator pin **must** rotate easily after the capscrews are tightened.

"Lock" the crankshaft in position. Make sure the locator pin is still in alignment and that the locator pin can be rotated easily.



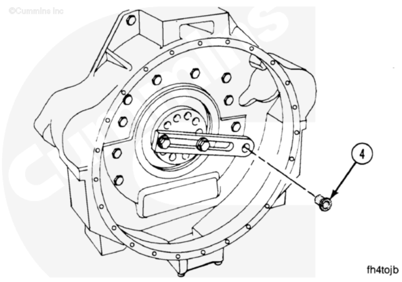
Measure the dowel pins to be installed.

Obtain a reamer (5) that is 0.013 mm to 0.025 mm [0.0005 inch to 0.001 inch] smaller than the dowel.

The dowel length **must** protrude from the block one-half of the flywheel housing wall thickness.

There are three oversize dowel pins available from Cummins Inc.

Oversize Dowel Pin Outside Diameter		
mm	**in**	**Oversize**
13.08	0.515	[0.015 in]
13.46	0.530	[0.030 in]
13.84	0.545	[0.045 in]



Install the appropriate drill bushings (4). The following bushings are available from Cummins Inc.

Drill/Ream Bushing Sets - 25.4 mm [1.0 inch]				
	Oversize mm	Oversize [Inch]	Bushing Size mm	Bushing Size [Inch]
3376495	Special		12.304	[0.4844]
	Standard		12.700	[0.5000]
	0.38	[0.015]	13.096	[0.5156]
	0.76	[0.030]	13.492	[0.5312]
	1.14	[0.045]	13.879	[0.5464]
ST-1234	Standard		14.288	[0.5625]

Drill/Ream Bushing Sets - 25.4 mm [1.0 inch]				
	0.38	[0.015]	14.684	[0.5781]
	0.76	[0.030]	15.080	[0.5937]
	1.14	[0.045]	15.479	[0.6094]
ST-1235	Standard		15.875	[0.6250]
	0.38	[0.015]	16.271	[0.6406]
	0.76	[0.030]	16.667	[0.6562]
	1.14	[0.045]	17.066	[0.6719]
ST-1236	Standard		17.463	[0.6875]
	0.38	[0.015]	17.859	[0.7031]
	0.76	[0.030]	18.255	[0.7187]
	1.14	[0.045]	18.654	[0.7344]
ST-1237	Standard		19.050	[0.7500]
	0.38	[0.015]	19.446	[0.7656]
	0.76	[0.030]	19.842	[0.7812]
ST-1238			22.621	[0.8906]
			23.813	[0.9375]

The drill bushing that is used **must** be the same size as the reamer (or the drill) that is used.

⚠ WARNING ⚠

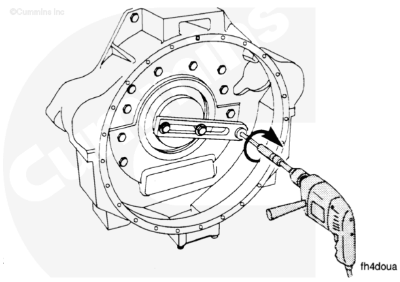
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the risk of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not allow metal chips to enter the engine. Damage to the engine will result.



When the new dowel pins are more than 0.38 mm [0.015 inch] larger than the old dowels, drill the hole to a size that is slightly smaller than the reamer. Then the reamer will **not** have to remove an excessive amount of material.

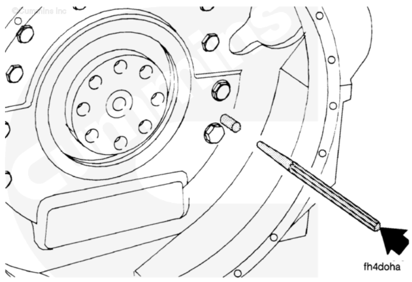
Ream the hole until the reamer touches the bottom of the hole in the block.

Remove the reamer. Clean the hole. Use solvent. Dry with compressed air. Push the reamer through the hole again. The reamer **must** touch the bottom of the hole in the block.

After reaming one hole, turn the plate and align it with the next dowel hole. Repeat the procedure in the next hole.

⚠ CAUTION ⚠

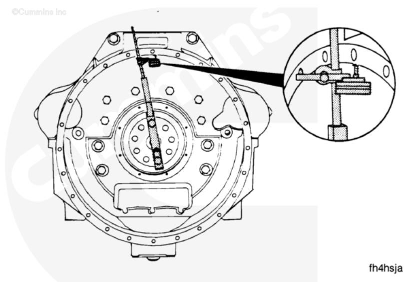
The dowel hole must not contain any metal chips. Damage will result.



Remove the plate from the crankshaft.

Use a square nose drift. Install each dowel until it touches the bottom of the hole in the block.

After the dowels are installed, measure the bore and the face alignment again.



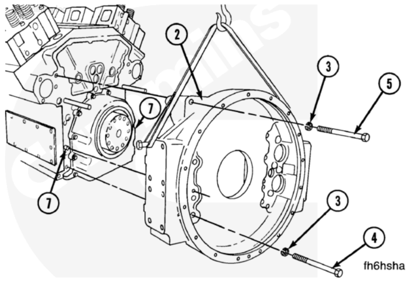
Install

Note : Newer engines have a flywheel housing that also contains a rear crankshaft seal. When this housing is used, the rear seal housing or the rear gear drive housing is not used. The newer housing is referred to as “Housing with Rear Seal” in the procedure. Older housings are referred to as “Housing without Rear Seal”.

Note : Guide bolts with 3/4-10 inch threads that are a minimum of 203 mm [8 in] in length will aid the assembly procedure.

⚠ WARNING ⚠

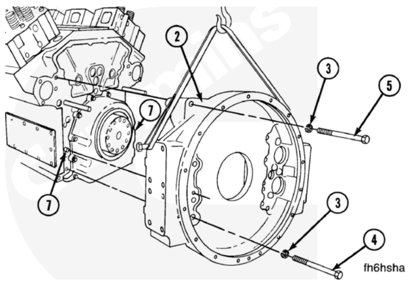
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Flywheel Housing without Rear Seal

- On flywheel housings without the rear seal, make sure the two dowel pins (7) are installed in the cylinder block.
- Check to make sure the mating surfaces of the block and the housing are clean, dry, and free from nicks and burrs.
- Install the flywheel housing (2). Push the housing over the dowel pins and to the cylinder block.
- The capscrews (4) and (5) **must** be SAE Grade 8.

Install the twelve lock washers (3) on the ten capscrews (4) and the two capscrews (5). Remove any guide bolts.

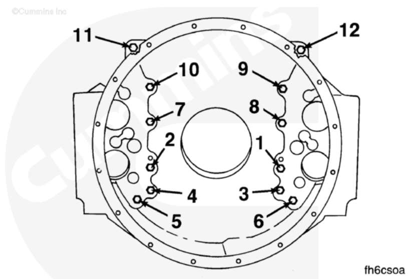


Follow the sequence shown to tighten the capscrews.

Torque Value:

¾ - 10 Inch Capscrews

1. 150 n•m [110 ft-lb]
2. 305 n•m [225 ft-lb]
3. 460 n•m [340 ft-lb]



⚠ WARNING ⚠

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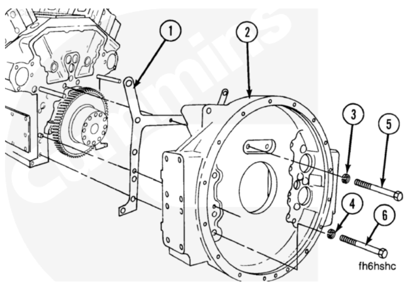
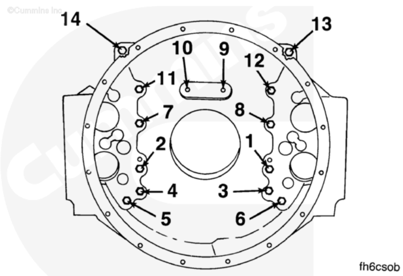


Flywheel Housing with Rear Seal

- On flywheel housings with the rear seal, make sure the two dowel pins are installed in the cylinder block.

- Check to make sure the mating surfaces of the block and the housing are clean, dry, and free from nicks and burrs.
- Install the gasket (1).
- Install the flywheel housing (2). Push the housing over the dowel pins and to the cylinder block.
- The capscrews (5) and (6) **must** be SAE Grade 8.
- The two capscrews (5) have 7/16-14 inch threads. The other capscrews (6) have 3/4-10 in threads.
- Install the two lock washers (3) and twelve lock washers (4).
- Install the two capscrews (5) and twelve capscrews (6).

Follow the sequence shown to tighten the capscrews.



Torque Value:
7/16-14 in Capscrews

1. 90 n•m [66 ft-lb]

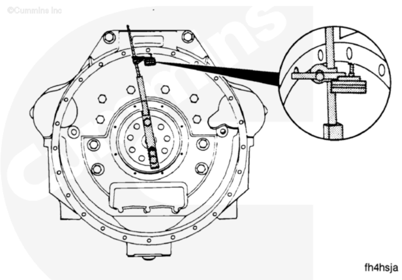
Torque Value:
¾ - 10 Inch Capscrews

1. 150 n•m [110 ft-lb]
2. 305 n•m [225 ft-lb]
3. 460 n•m [340 ft-lb]

Measure

Alignment

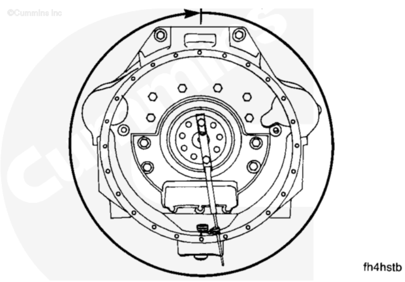
The bore and the face of the housing **must** be in alignment with the crankshaft.



The indicator arm **must** be rigid for an accurate reading. It **must not** sag.

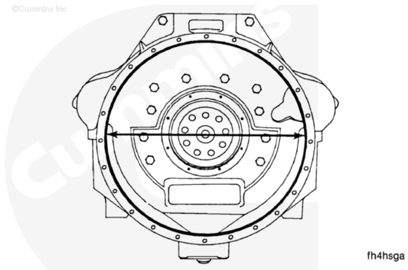
Attach an indicator to the crankshaft as shown to check the radial run out.

Position the indicator at the 12 o'clock position. Adjust the dial until the needle points to “0”.



Rotate the crankshaft 1 complete revolution (360 degrees).

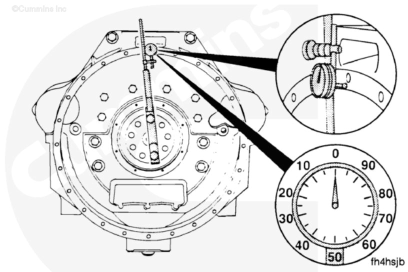
Record the total indicator runout.



The maximum allowable total indicator runout depends on the diameter of the bore.

Bore Diameter			Maximum	
MIN/MAX	MIN/MAX	SAE Number	Total Indicator Runout	
mm	in		mm	in
787.4/810.5	[31.00/31.91]	00	0.30	0.012
647.7/648.0	[25.50/25.51]	0	0.25	0.010

If the alignment is **not** within specifications and the bore is round, the housing can be shifted. If the alignment is **not** within specifications and the bore is **not** round, the housing **must** be replaced.

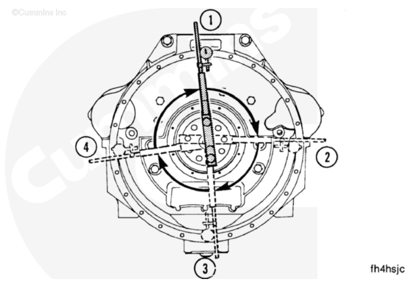


⚠ CAUTION ⚠

The crankshaft must be pushed or pulled in the same direction each time it is measured.

Face Runout

Attach an indicator at the 12 o'clock position. Adjust the dial until the needle points to “0”.



Record the indicator reading at three different points: 3 o'clock, 6 o'clock, and 9 o'clock.

Turn to the original 12 o'clock position. Make sure the needle still points to “0”.

Determine the total indicator runout.

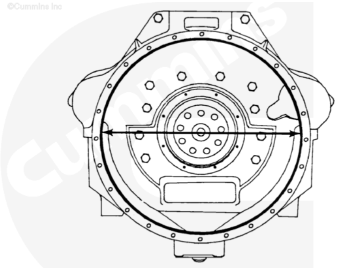
Total Indicator Runout		
Example	**mm**	**in**
12 o'clock	0.00	[0.00]
3 o'clock	+0.08	[+0.003]
6 o'clock	-0.05	[-0.002]
9 o'clock	+0.08	[+0.003]
Equal Total Indicator Runout	0.13	[0.005]

The maximum allowable total indicator runout depends on the diameter of the bore.

Bore Diameter			Maximum	
MIN/MAX	MIN/MAX	SAE Number	Total Indicator Runout	
mm	in		mm	in
787.4/810.5	[31.00/31.91]	00	0.30	0.012
647.7/648.0	[25.50/25.51]	0	0.25	0.010

If the alignment is **not** within specifications, remove the housing. Check for nicks, burrs, or foreign material between the block and the housing.

Check the alignment again. If the alignment is **not** within specifications, the block or the housing is **not** machined correctly.



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Finishing Steps

⚠ WARNING ⚠

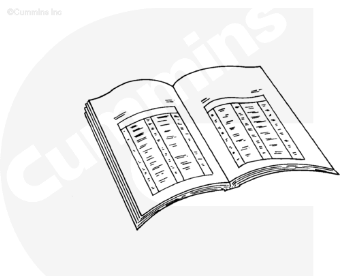
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged



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contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Install the rear crankshaft seal in the flywheel housing with rear seal. Refer to Procedure 001-024 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-024-tr.html)
- Install the oil pan adapter assembly. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)
- Install the starter motor. Refer to Procedure 013-020 in Section 13. (/qs3/pubsys2/xml/en/procedures/28/28-013-020-tr.html)
- Install the transmission, clutch, and all related components. See the equipment manufacturer service information.
- On QSK50 engines, equipped with electronically actuated injectors, install the engine speed sensor, use the special socket tool, Part Number 4918623, or equivalent. Refer to Procedure 019-042 in the Troubleshooting and Repair Manual Electronic Control System, QSK50 and QSK60 Modular Common Rail System, Bulletin 4021533. (/qs3/pubsys2/xml/en/procedures/05/05-019-042.html)
- Install new full flow oil filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)
- Fill the engine with oil. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Connect the air supply to the air starter. Refer to Procedure 012-022 in Section 12. (/qs3/pubsys2/xml/en/procedures/102/102-012-022.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to operating temperature. Check for leaks. Check equipment operation.